

# DBMS – SELLER AND SHOP MANAGEMENT

Project Title: Etsy Marketplace – Handmade and Vintage Products E-Commerce System

Skill Task: DBMS

Week: 3

## 1. Database Design

### 1.1 Seller Table

| Attribute    | Data Type     | Constraint       |
|--------------|---------------|------------------|
| Seller_ID    | INT           | Primary Key      |
| Seller_Name  | VARCHAR2(100) | NOT NULL         |
| Shop_Name    | VARCHAR2(100) | NOT NULL         |
| Email        | VARCHAR2(100) | NOT NULL, UNIQUE |
| Password     | VARCHAR2(50)  | NOT NULL         |
| Phone_Number | VARCHAR2(15)  | NOT NULL, UNIQUE |
| Address      | VARCHAR2(200) | NOT NULL         |
| Rating       | NUMBER(2,1)   | –                |
| Join_Date    | DATE          | NOT NULL         |

### 1.2 Shop Table

| Attribute   | Data Type     | Constraint          |
|-------------|---------------|---------------------|
| Shop_ID     | INT           | Primary Key         |
| Seller_ID   | INT           | Foreign Key, UNIQUE |
| Shop_Name   | VARCHAR2(100) | NOT NULL            |
| Description | VARCHAR2(255) | –                   |
| Logo        | VARCHAR2(200) | –                   |
| Location    | VARCHAR2(100) | NOT NULL            |

## 2. Table Relationships

| Parent Entity | Relationship | Child Entity | Explanation                          |
|---------------|--------------|--------------|--------------------------------------|
| Seller        | One-to-One   | Shop         | One seller owns exactly one shop.    |
| Seller        | One-to-Many  | Product      | One seller can list many products.   |
| Category      | One-to-Many  | Product      | One category contains many products. |

### 3. Table Creation

```
SQL> CREATE TABLE Seller (  
2     Seller_ID      INT PRIMARY KEY,  
3     Seller_Name    VARCHAR2(100) NOT NULL,  
4     Shop_Name      VARCHAR2(100) NOT NULL,  
5     Email          VARCHAR2(100) NOT NULL UNIQUE,  
6     Password       VARCHAR2(50)  NOT NULL,  
7     Phone_Number   VARCHAR2(15)  NOT NULL UNIQUE,  
8     Address        VARCHAR2(200) NOT NULL,  
9     Rating         NUMBER(2,1),  
10    Join_Date      DATE NOT NULL  
11 );
```

#### 3.2 Create Shop Table

```
SQL> CREATE TABLE Shop (  
2     Shop_ID        INT PRIMARY KEY,  
3     Seller_ID      INT NOT NULL UNIQUE,  
4     Shop_Name      VARCHAR2(100) NOT NULL,  
5     Description    VARCHAR2(255),  
6     Logo           VARCHAR2(200),  
7     Location       VARCHAR2(100) NOT NULL,  
8     FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID)  
9 );
```

#### 3.3 Create Category Table

```
SQL> CREATE TABLE Category (  
2     Category_ID    INT PRIMARY KEY,  
3     Category_Name  VARCHAR2(100) NOT NULL,  
4     Description    VARCHAR2(255)  
5 );
```

Table created.

#### 3.4 Create Product Table

```
SQL> CREATE TABLE Product (  
2     Product_ID     INT PRIMARY KEY,  
3     Seller_ID      INT NOT NULL,  
4     Category_ID    INT NOT NULL,  
5     Product_Name   VARCHAR2(100) NOT NULL,  
6     Description    VARCHAR2(255),  
7     Price          NUMBER(10,2) NOT NULL,  
8     Quantity       INT NOT NULL,  
9     Material       VARCHAR2(50),  
10    Color          VARCHAR2(30),  
11    Product_Size   VARCHAR2(20),  
12    Image_URL      VARCHAR2(200),  
13    Listing_Date   DATE NOT NULL,  
14    FOREIGN KEY (Seller_ID) REFERENCES Seller(Seller_ID),  
15    FOREIGN KEY (Category_ID) REFERENCES Category(Category_ID)  
16 );
```

Table created.

## 4. Seller Data Insertion

```
SQL> INSERT INTO Seller VALUES (501, 'Aarthi Handmade Crafts', 'Handmade Haven', 'aarthi.crafts@gmail.com', 'Pass@123', '9000000001', 'Coimbatore', 4.5, '10-JAN-2026');
1 row created.

SQL> INSERT INTO Seller VALUES (502, 'Rahul Vintage Store', 'Vintage Vault', 'rahul.vintage@gmail.com', 'Pass@456', '9000000002', 'Chennai', 4.2, '15-FEB-2026');
1 row created.

SQL> INSERT INTO Seller VALUES (503, 'Meera Jewel Works', 'Bead and Bloom', 'meera.jewels@gmail.com', 'Pass@789', '9000000003', 'Madurai', 4.8, '02-MAR-2026');
1 row created.

SQL> INSERT INTO Seller VALUES (504, 'Karthik Wood Art', 'Woodcraft Studio', 'karthik.wood@gmail.com', 'Pass@321', '9000000004', 'Bangalore', 4.0, '20-MAR-2026');
1 row created.

SQL> INSERT INTO Seller VALUES (505, 'Divya Home Decor', 'Decor Diaries', 'divya.decor@gmail.com', 'Pass@654', '9000000005', 'Salem', 3.9, '05-APR-2026');
1 row created.
```

## 5. Shop Data Insertion

The Shop rows and the Category rows are inserted in the same session.

```
1 row created.

SQL> INSERT INTO Shop VALUES (601, 501, 'Handmade Haven', 'Handmade cotton and jute bags', 'handmade_haven.png', 'Coimbatore');
1 row created.

SQL> INSERT INTO Shop VALUES (602, 502, 'Vintage Vault', 'Vintage collectibles and antique decor', 'vintage_vault.png', 'Chennai');
1 row created.

SQL> INSERT INTO Shop VALUES (603, 503, 'Bead and Bloom', 'Handcrafted beaded jewellery', 'bead_bloom.png', 'Madurai');
1 row created.

SQL> INSERT INTO Shop VALUES (604, 504, 'Woodcraft Studio', 'Wooden toys and carved showpieces', 'woodcraft_studio.png', 'Bangalore');
1 row created.

SQL> INSERT INTO Shop VALUES (605, 505, 'Decor Diaries', 'Home decor and wall hangings', 'decor_diaries.png', 'Salem');
1 row created.

SQL> INSERT INTO Category VALUES (1, 'Handmade Bags', 'Cloth and jute bags');
1 row created.

SQL> INSERT INTO Category VALUES (2, 'Jewellery', 'Handcrafted jewellery');
1 row created.

SQL> INSERT INTO Category VALUES (3, 'Home Decor', 'Wall and table decor');
1 row created.
```

### 5.1 Product Data Insertion

```
SQL> INSERT INTO Product VALUES (701, 501, 1, 'Handmade Jute Tote Bag', 'Eco friendly jute bag', 450, 40, 'Jute', 'Brown', 'Medium', 'jute_tote.jpg', '12-JAN-2026');
1 row created.

SQL> INSERT INTO Product VALUES (702, 503, 2, 'Beaded Silver Necklace', 'Handcrafted bead necklace', 1200, 25, 'Silver', 'Silver', 'Free', 'bead_necklace.jpg', '05-MAR-2026');
1 row created.

SQL> INSERT INTO Product VALUES (703, 504, 3, 'Carved Wooden Elephant', 'Rosewood carved showpiece', 850, 0, 'Wood', 'Brown', 'Small', 'wood_elephant.jpg', '22-MAR-2026');
1 row created.

SQL> INSERT INTO Product VALUES (704, 502, 4, 'Vintage Brass Lamp', 'Antique brass table lamp', 2100, 10, 'Brass', 'Golden', 'Medium', 'brass_lamp.jpg', '18-FEB-2026');
1 row created.

SQL> INSERT INTO Product VALUES (705, 505, 3, 'Macrame Wall Hanging', 'Cotton macrame wall decor', 700, 15, 'Cotton', 'White', 'Large', 'macrame.jpg', '08-APR-2026');
1 row created.
```

## 6. Seller Shop Information

### 6.1 Display Seller and Shop Details

This query combines Seller and Shop information.

```
SQL> SELECT
  2     s.Seller_ID,
  3     s.Seller_Name,
  4     sh.Shop_Name,
  5     sh.Location,
  6     s.Rating
  7 FROM Seller s
  8 JOIN Shop sh
  9 ON s.Seller_ID = sh.Seller_ID
 10 ORDER BY s.Seller_Name;
```

| SELLER_ID | SELLER_NAME            | SHOP_NAME        | LOCATION   | RATING |
|-----------|------------------------|------------------|------------|--------|
| 501       | Aarthi Handmade Crafts | Handmade Haven   | Coimbatore | 4.5    |
| 505       | Divya Home Decor       | Decor Diaries    | Salem      | 3.9    |
| 504       | Karthik Wood Art       | Woodcraft Studio | Bangalore  | 4      |
| 503       | Meera Jewel Works      | Bead and Bloom   | Madurai    | 4.8    |
| 502       | Rahul Vintage Store    | Vintage Vault    | Chennai    | 4.2    |

5 rows selected.

### 6.2 Display Seller, Shop and Product Details

This query combines Seller, Shop and Product information.

```
SQL> SELECT
  2     s.Seller_Name,
  3     sh.Shop_Name,
  4     p.Product_Name,
  5     p.Price,
  6     p.Quantity
  7 FROM Seller s
  8 JOIN Shop sh
  9 ON s.Seller_ID = sh.Seller_ID
 10 JOIN Product p
 11 ON s.Seller_ID = p.Seller_ID
 12 ORDER BY s.Seller_Name;
```

| SELLER_NAME            | SHOP_NAME        | PRODUCT_NAME           | PRICE | QUANTITY |
|------------------------|------------------|------------------------|-------|----------|
| Aarthi Handmade Crafts | Handmade Haven   | Handmade Jute Tote Bag | 450   | 40       |
| Divya Home Decor       | Decor Diaries    | Macrame Wall Hanging   | 700   | 15       |
| Karthik Wood Art       | Woodcraft Studio | Carved Wooden Elephant | 850   | 0        |
| Meera Jewel Works      | Bead and Bloom   | Beaded Silver Necklace | 1200  | 25       |
| Rahul Vintage Store    | Vintage Vault    | Vintage Brass Lamp     | 2100  | 10       |

5 rows selected.

## 7. Product Stock Status

### 7.1 Display Available Products

```
SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     p.Quantity,
  5     'Available' AS Stock_Status
  6 FROM Product p
  7 WHERE p.Quantity > 0;
```

| PRODUCT_ID | PRODUCT_NAME | QUANTITY | STOCK_STATUS |
|------------|--------------|----------|--------------|
|------------|--------------|----------|--------------|

```

-----
701 Handmade Jute Tote Bag      40 Available
702 Beaded Silver Necklace    25 Available
704 Vintage Brass Lamp        10 Available
705 Macrame Wall Hanging      15 Available
-----

```

4 rows selected.

## 7.2 Display Unavailable Products

```

SQL> SELECT
  2     p.Product_ID,
  3     p.Product_Name,
  4     p.Quantity,
  5     'Unavailable' AS Stock_Status
  6 FROM Product p
  7 WHERE p.Quantity = 0;

```

```

PRODUCT_ID  PRODUCT_NAME          QUANTITY  STOCK_STATUS
-----
703 Carved Wooden Elephant      0 Unavailable

```

1 row selected.

## 8. Stock Update

Demonstrate updating the stock information of a product.

*Note: Product\_ID 703 (Carved Wooden Elephant) is intentionally left at zero stock so the zero-stock report in Section 9.4 still has a row to show. The restock below is applied to Product\_ID 705 (Macrame Wall Hanging) instead.*

```

SQL> SELECT Product_ID, Product_Name, Quantity
  2 FROM Product
  3 WHERE Quantity = 0;

```

```

PRODUCT_ID  PRODUCT_NAME          QUANTITY
-----
703 Carved Wooden Elephant      0

```

1 row selected.

```

SQL> UPDATE Product
  2 SET Quantity = 30
  3 WHERE Product_ID = 705;

```

1 row updated.

```

SQL> SELECT Product_ID, Product_Name, Quantity
  2 FROM Product
  3 WHERE Product_ID = 705;

```

```

PRODUCT_ID  PRODUCT_NAME          QUANTITY
-----
705 Macrame Wall Hanging      30

```

1 row selected.

## 9. Seller and Stock Reports

### 9.1 Complete Seller – Shop – Product Report

```

SQL> SELECT
  2     s.Seller_Name,
  3     sh.Shop_Name,
  4     c.Category_Name,
  5     p.Product_Name,

```

```

6      p.Price,
7      p.Quantity
8  FROM Product p
9  JOIN Seller s
10 ON p.Seller_ID = s.Seller_ID
11 JOIN Shop sh
12 ON s.Seller_ID = sh.Seller_ID
13 JOIN Category c
14 ON p.Category_ID = c.Category_ID
15 ORDER BY s.Seller_Name;

```

| SELLER_NAME            | SHOP_NAME        | CATEGORY_NAME | PRODUCT_NAME           | PRICE | QUANTITY |
|------------------------|------------------|---------------|------------------------|-------|----------|
| Aarthi Handmade Crafts | Handmade Haven   | Handmade Bags | Handmade Jute Tote Bag | 450   | 40       |
| Divya Home Decor       | Decor Diaries    | Home Decor    | Macrame Wall Hanging   | 700   | 30       |
| Karthik Wood Art       | Woodcraft Studio | Home Decor    | Carved Wooden Elephant | 850   | 0        |
| Meera Jewel Works      | Bead and Bloom   | Jewellery     | Beaded Silver Necklace | 1200  | 25       |
| Rahul Vintage Store    | Vintage Vault    | Vintage       | Vintage Brass Lamp     | 2100  | 10       |

5 rows selected.

## 9.2 Available and Unavailable Product Count

```

SQL> SELECT
2      CASE WHEN Quantity > 0 THEN 'Available'
3      ELSE 'Unavailable' END AS Stock_Status,
4      COUNT(*) AS Total_Products
5  FROM Product
6  GROUP BY CASE WHEN Quantity > 0 THEN 'Available'
7      ELSE 'Unavailable' END;

```

| STOCK_STATUS | TOTAL_PRODUCTS |
|--------------|----------------|
| Available    | 4              |
| Unavailable  | 1              |

2 rows selected.

## 9.3 Total Stock by Seller

```

SQL> SELECT
2      s.Seller_Name,
3      SUM(p.Quantity) AS Total_Stock
4  FROM Seller s
5  JOIN Product p
6  ON s.Seller_ID = p.Seller_ID
7  GROUP BY s.Seller_Name
8  ORDER BY s.Seller_Name;

```

| SELLER_NAME            | TOTAL_STOCK |
|------------------------|-------------|
| Aarthi Handmade Crafts | 40          |
| Divya Home Decor       | 30          |
| Karthik Wood Art       | 0           |
| Meera Jewel Works      | 25          |
| Rahul Vintage Store    | 10          |

5 rows selected.

## 9.4 Products with Zero Stock

This is particularly useful for demonstrating unavailable products.

```

SQL> SELECT
2      p.Product_ID,
3      p.Product_Name,
4      s.Seller_Name,
5      sh.Shop_Name,

```

```

6      p.Quantity
7 FROM Product p
8 JOIN Seller s
9 ON p.Seller_ID = s.Seller_ID
10 JOIN Shop sh
11 ON s.Seller_ID = sh.Seller_ID
12 WHERE p.Quantity = 0;

```

| PRODUCT_ID | PRODUCT_NAME           | SELLER_NAME      | SHOP_NAME        | QUANTITY |
|------------|------------------------|------------------|------------------|----------|
| 703        | Carved Wooden Elephant | Karthik Wood Art | Woodcraft Studio | 0        |

1 row selected.

## 9.5 Highest Rated Sellers

```

SQL> SELECT
2      s.Seller_Name,
3      sh.Shop_Name,
4      s.Rating
5 FROM Seller s
6 JOIN Shop sh
7 ON s.Seller_ID = sh.Seller_ID
8 WHERE s.Rating >= 4.5
9 ORDER BY s.Rating DESC;

```

| SELLER_NAME            | SHOP_NAME      | RATING |
|------------------------|----------------|--------|
| Meera Jewel Works      | Bead and Bloom | 4.8    |
| Aarthi Handmade Crafts | Handmade Haven | 4.5    |

2 rows selected.

## 10. Conclusion

The Seller and Shop modules of the Etsy Marketplace Database System were created, populated with sample data, and queried successfully. The One-to-One relationship between Seller and Shop and the One-to-Many relationship between Seller and Product were enforced using primary key, unique and foreign key constraints. Join queries, aggregate reports and stock updates confirm that the design maintains data integrity and supports the reporting needs of the marketplace.